UNIVERSITÉ CÔTE D'AZUR | MASTER 2 – SLCD

Big Data and NoSQL

Project 1 – WeatherScope

A Historical Weather Observatory

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Groups	Two students. All members must be able to explain any part of the submitted work.
Submission	**Friday, 6 November 2026**
Defence	**Tuesday, 10 November 2026, at 2:00 p.m.**
Deliverables	A **written report** (at most 10 pages, excluding appendices); the **source code**; a **15-minute oral presentation** including a live demonstration, followed by a **10-minute question-and-answer session** with a live challenge.

1. Project scenario

A regional environmental observatory wants to investigate unusual weather conditions using historical measurements from monitoring stations. Its analysts need to identify periods of exceptional heat or precipitation, compare the behaviour of different areas, and examine the reliability of the observations behind a result.

You are commissioned to develop **WeatherScope**, a working prototype through which an analyst can select a location and a period, investigate weather events, compare areas and obtain supporting measurements. The observatory expects the platform to remain usable as more stations and years of records are incorporated, and it will only trust a result if it can be traced back to the original observations.

The prototype will first be deployed using a **single local data server**. It will then be deployed using **several data-serving instances** on the same workstation to study how distributing data changes ingestion and query performance. Both deployments must provide the same functionality and return identical analytical results.

2. Data

Use the **Global Historical Climatology Network–Daily (GHCN-Daily)** provided by the US National Oceanic and Atmospheric Administration (NOAA). This archive contains recorded daily weather observations and information about the stations that produced them.

Resource	Official source
Observations by year	https://www.ncei.noaa.gov/pub/data/ghcn/daily/by_year/
Observations by station	https://www.ncei.noaa.gov/pub/data/ghcn/daily/by_station/
Station metadata	https://www.ncei.noaa.gov/pub/data/ghcn/daily/ghcnd-stations.txt
Format and conventions	https://www.ncei.noaa.gov/pub/data/ghcn/daily/readme.txt
CSV format	https://www.ncei.noaa.gov/pub/data/ghcn/daily/readme-by_year.txt

Each observation contains a station identifier, a date, a meteorological variable (*element*), its recorded value, a measurement flag, a quality flag, a source flag and, for some records, an observation time.

Station metadata provide names, coordinates and elevation. Useful elements include daily maximum temperature (`TMAX`), daily minimum temperature (`TMIN`) and precipitation (`PRCP`). In the source files, `TMAX` and `TMIN` are expressed in **tenths of a degree Celsius** and `PRCP` in **tenths of a millimetre**. Each yearly file covers all stations worldwide; the per-station files are usually more convenient for a limited set of stations. Many stations record their daily values at a fixed morning or evening observation time, so a calendar date does not cover the same 24 hours at every station; discuss the consequences of this for multi-station comparisons.

Scope and registration

Choose a documented geographical scope covering **at least two distinct areas** and **at least eight consecutive calendar years**. The notion of *area* (bounding box, country, state, radius around a point, ...) is yours to define and justify. The selected subset must contain **at least 100 stations and 200,000 distinct station–day pairs** (a station–day counts if at least one of `TMAX`, `TMIN` or `PRCP` is recorded for it); any adjustment must be supported by the measured availability of observations or by material constraints.

Each group registers its scope (areas, years, intended reference period) with the instructor before starting the implementation. **No two groups may register the same pair of areas**; the instructor may ask a group to modify a scope that overlaps too much with another one. The instructor will use the registered scopes to prepare the defence challenges (Section 6).

Provenance and quality

The archive is revised continuously: the same file downloaded on two different dates may differ. Record the retrieval date, the exact list of source files and a checksum of each of them. Report the number of stations, station–day pairs and usable observations for each analysed element, and characterise temporal coverage, geographical distribution and missingness.

In an appendix, document **at least five data anomalies actually present in your subset** (for example: flagged values, physically implausible values, inconsistent `TMAX`/`TMIN` pairs, gaps, stations whose record starts or stops during the period), each with its frequency and at least one concrete station and date. These examples will be checked against the NOAA files.

The application and benchmarks must use **real observations only**. Do not manufacture measurements, duplicate records to inflate the dataset or count replicated copies as new observations. Explain how incomplete histories, duplicate source records, unusual values, quality flags and measurement flags are treated. In particular, precipitation values carrying the measurement flag `P` (missing, presumed zero) or `T` (trace) must not be used as ordinary zeros without justification.

3. System to develop

WeatherScope must expose the following services through a usable interface (web interface, API or command-line application). Query parameters must be supplied at execution time, and results must be computed from the stored data. Precomputed aggregates or materialised views are allowed if the system maintains them and F5 keeps them up to date; answers prepared in advance for fixed examples, or tables assembled by hand, are not.

F1. Explore a station. Search for stations within a geographical area, select a station and retrieve its measurements over a chosen date interval. Present a time-based summary together with the availability and quality of the underlying observations.

F2. Investigate prolonged events. Given a geographical area, a period and a user-defined event criterion, identify episodes of consecutive days characterised by exceptional heat or precipitation.

Return the stations concerned, the start and end dates, the event duration and a measure of its intensity. Missing days and observations with quality concerns must not silently create or extend episodes, and **an episode crossing a year boundary must be reported as a single episode**.

F3. Compare historical conditions. For selected areas and years, identify months that depart most strongly from the usual seasonal behaviour in temperature or precipitation. Produce a ranked comparison that accounts for differences in station coverage and observation availability. Define and justify the historical reference and the eligibility of incomplete periods. The reference period may extend beyond the study period; observations used only as a reference must be loaded through the same pipeline and counted separately in the report. Climatological normals are conventionally computed over 30 years: if your reference period is much shorter, discuss the statistical consequences.

F4. Identify regional events. Find periods during which several stations within an area experience unusual conditions over overlapping dates. Distinguish an isolated extreme at one station from an event affecting a wider geographical area, taking into account that station density is not uniform. Return evidence supporting the detected events and their relative severity.

F5. Update the archive. Import an additional batch of genuine NOAA observations (for instance a year or a set of stations excluded from the initial load) without losing previous data or creating duplicates. Define the identity of an observation and a justified policy for records that already exist with a *different* value or flag (overwrite, keep a history, or reject). Importing the same batch twice must leave the archive unchanged. Genuine revisions only appear between two downloads made at different dates: the policy must be implemented, but testing it on real revised records is optional. The update must work on both deployments, and the services above must reflect the updated archive.

For each service, document its inputs and outputs and give at least one meaningful example obtained from the selected dataset. Any reported event must be traceable to its source observations: given a result, the system must be able to list the underlying records (station, date, element, value, flags).

4. Implementation and distribution

Part I – Local prototype

Design the data organisation and implement all five services on a single local data server, that is, a database system running as a separate server process queried by the application through a network endpoint (an embedded library such as SQLite or DuckDB qualifies only if it is wrapped in such a server process). The report should present the structure of the stored information, the reasons for the data-management technology selected, the main access patterns and the mechanisms used to make the required operations efficient.

Use actual execution results to justify at least one important design decision. Implement and measure **one alternative design** (data model, indexing or physical organisation) on the same data and on at least two representative requests, and explain why it was retained or rejected. The full acquisition and loading procedure must be reproducible.

Part II – Distributed prototype

Deploy a second version using **at least two separate data-serving instances**, running as processes or containers with distinct storage locations and communication endpoints. A data-serving instance is a database or storage server that stores part of the observations and answers queries on them; a processing engine alone (for instance Spark reading files) does not count as one. Observations must

be placed across the instances, rather than merely copied in full to each of them. WeatherScope must continue to expose a single coherent interface to its users.

Define and justify the data-placement strategy. For representative requests, show which instances are contacted and how partial results are combined. Contrast requests with narrow geographical and temporal scopes with requests involving broad areas or long histories. Evaluate the effect of the chosen organisation on data balance, writes and multi-station analyses. Verify that the local and distributed versions produce **identical answers** for the same observations and requests, using an automated comparison. Compare results after a canonical ordering, with ties broken deterministically. The comparison must include requests whose time range or set of stations spans several instances.

Implement and evaluate **one alternative distributed configuration**, changing either the placement policy or the number of data-serving instances.

Losing an instance. Stop one data-serving instance and run F1 on a station stored on that instance, then F3. Show what the user receives. A partial answer must never be presented as complete: either the request fails explicitly, or the response states which data were unavailable. Distinguish what your implementation actually recovers from what a replicated production system could recover. A multi-process deployment on one workstation must not be presented as equivalent to a cluster of independent machines.

5. Experimental investigation

The observatory will use WeatherScope in two different ways: routine examination of a limited number of stations over short periods, and occasional regional studies spanning many stations and years.

E1. Increasing data volume. Evaluate ingestion and representative service requests on three increasingly large subsets of the *real* archive. The largest subset must contain **at least 10 million raw observations** of `TMAX`, `TMIN` and `PRCP`, obtained from additional real stations or years (the per-year files may be needed); observations loaded for the reference period count towards this total. State whether the largest subset still fits in the memory available to your database. Record the number of observations, execution times and storage requirements.

E2. Local versus distributed operation. Compare the local deployment and both distributed configurations on identical data and service requests, using the largest subset of E1 (E3 uses the same data). Include a narrow query, a long-range aggregation and an analysis involving multiple stations. Report repeated measurements with central tendency and variability, and state the cache and warm-up conditions.

E3. Changing workload. Construct two workloads that reflect the observatory's routine and regional-study modes. Choose at least two of your three configurations to compare. **Before running the measurements**, write down which one you expect to perform better under each workload and why, and deposit this prediction on the course platform (or send it to the instructor). Vary the relative frequency of the two request types and confront the predictions with the measured results. A wrong prediction that is well explained is valued as much as a correct one.

E4. A limitation of your design (optional). This experiment is optional and counts as a bonus. Find a case, drawn from the *actual* geographical or temporal distribution of the observations, where the chosen architecture performs poorly or yields a potentially misleading result. Determine the cause, make one well-motivated change and measure its effect. Discuss any improvement obtained at the expense of other operations or resources.

The results must include the hardware and software versions, resource limits, query parameters, timings, relevant instance-level observations and an explanation of the differences. On a single work-

station, several data-serving processes still share CPU, memory and disk; account for this when interpreting the impact of distribution, and do not assume that adding instances improves performance.

Independent verification. Verify at least **two non-trivial analytical results** (one from F2 and one from F3 or F4) with a separate script that reads the original NOAA files directly, without using your database.

6. Submission and defence

Submission package

By **Friday, 6 November 2026**, submit: the written report (at most 10 pages, excluding appendices) presenting the system, architecture, verified results, experiments and limitations; the complete source code and configuration files; the raw measurements used in the comparisons; and a README with setup and execution instructions allowing the main experiments to be reproduced.

Defence

Each group gives a **15-minute presentation**, including a live demonstration of at most five minutes, followed by a **10-minute question-and-answer session**. During the questions, the group will face a **live challenge** prepared from its registered scope, for example:

- running services with new parameters chosen by the jury;
- importing, through F5, a genuine NOAA extract supplied at the defence, then showing that the results change accordingly;
- tracing a reported result back to its source records;
- explaining a file or function of the source code chosen by the jury, by a group member also chosen by the jury.